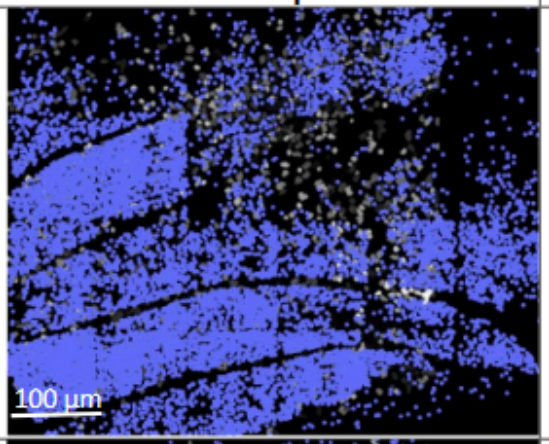
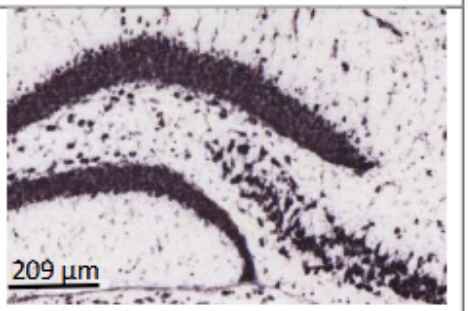
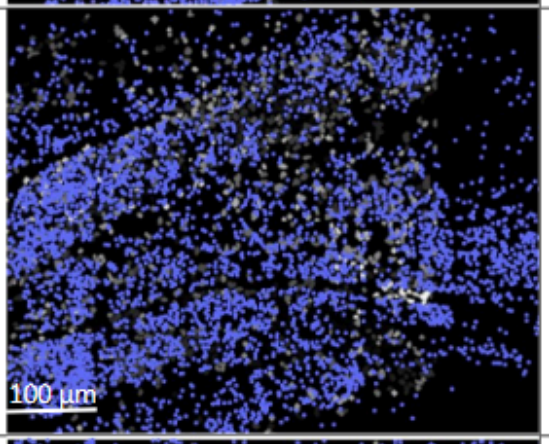
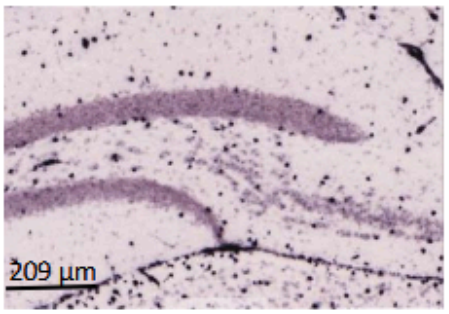
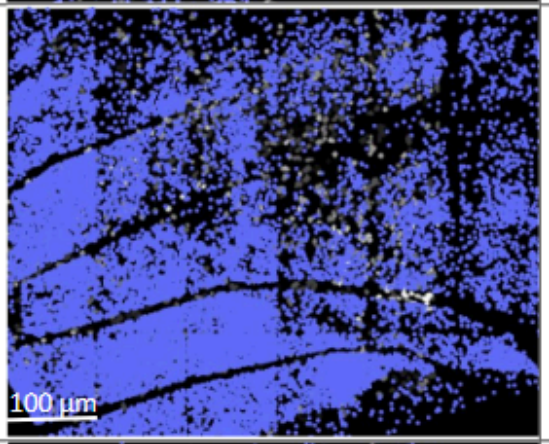
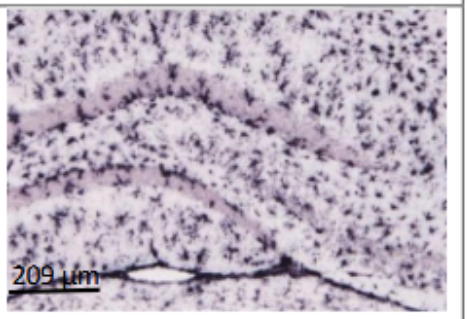
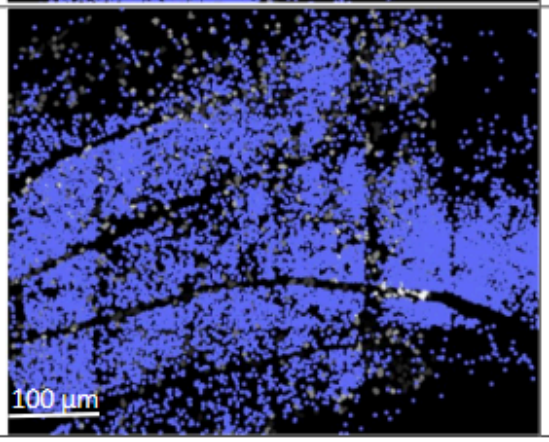
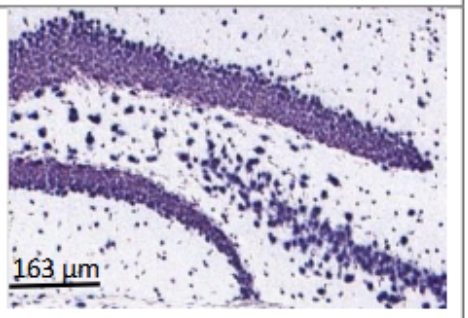
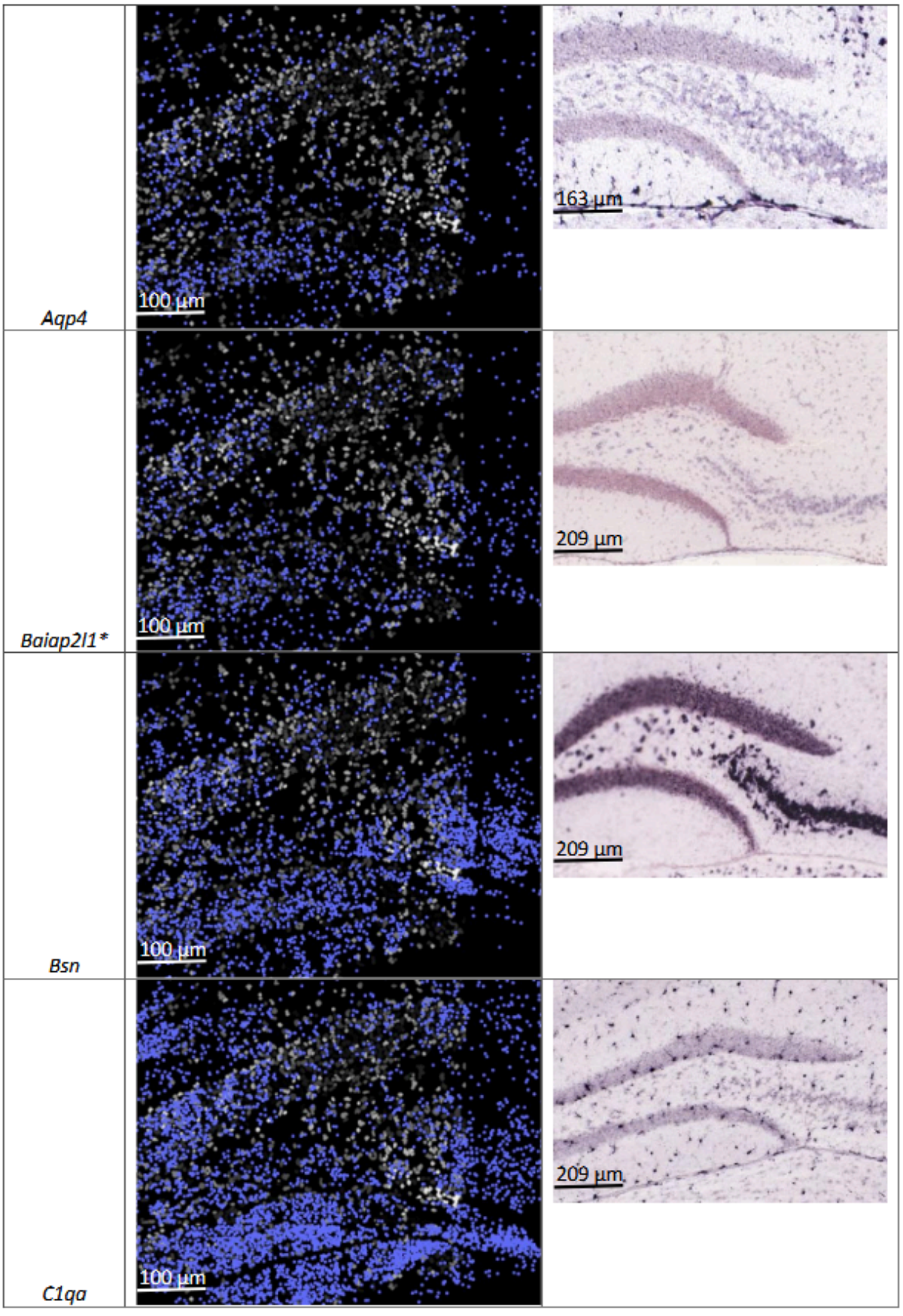
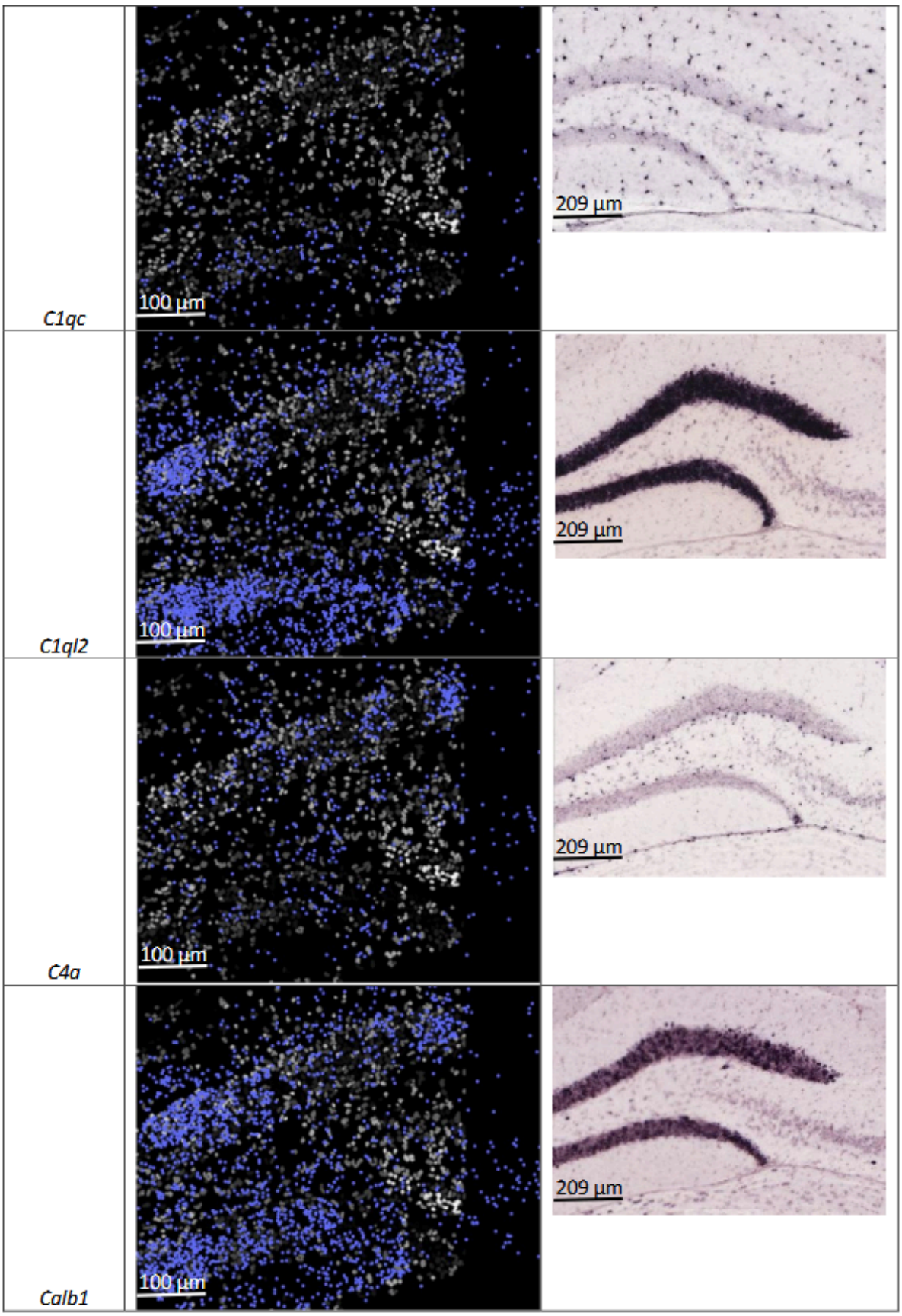
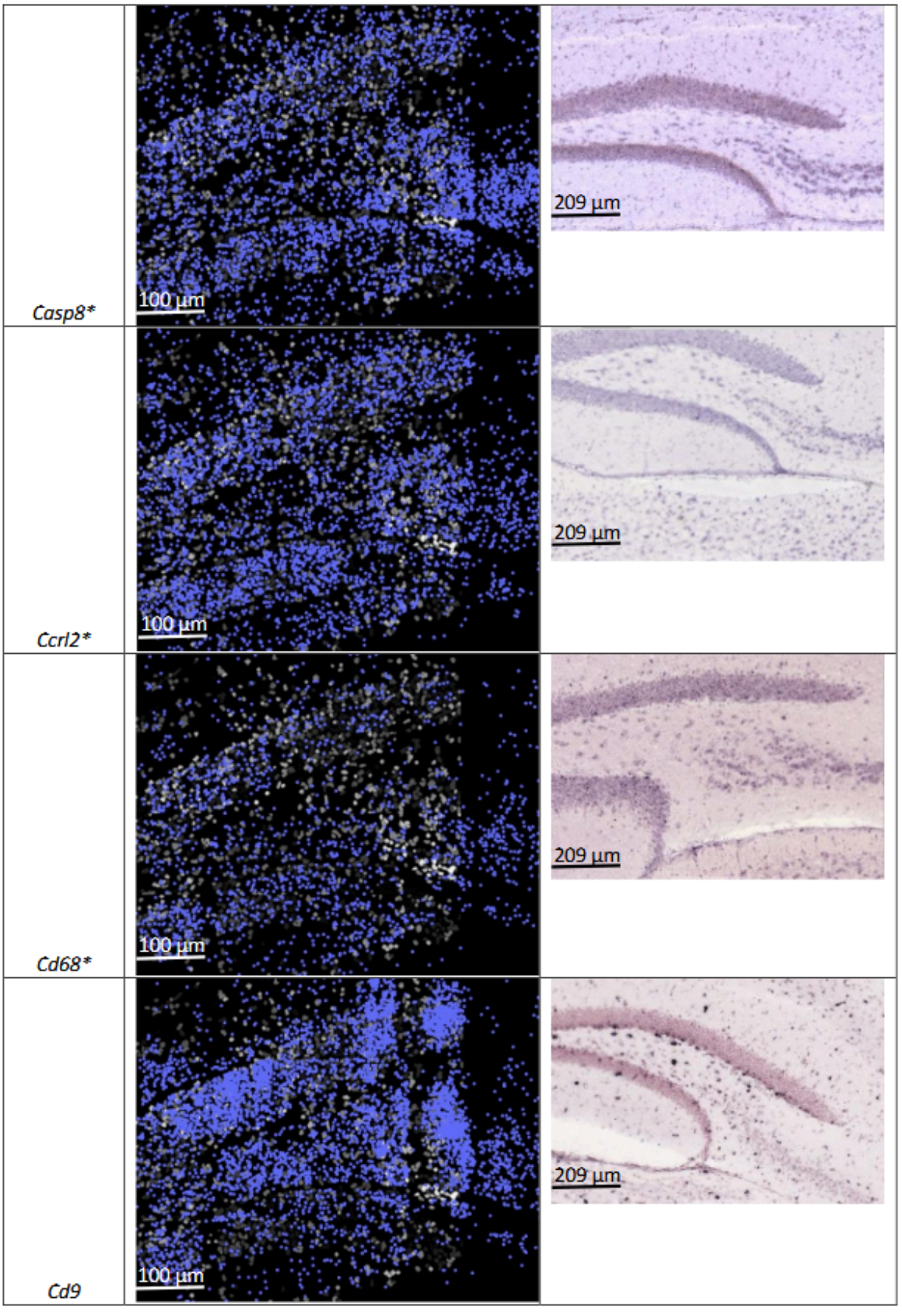
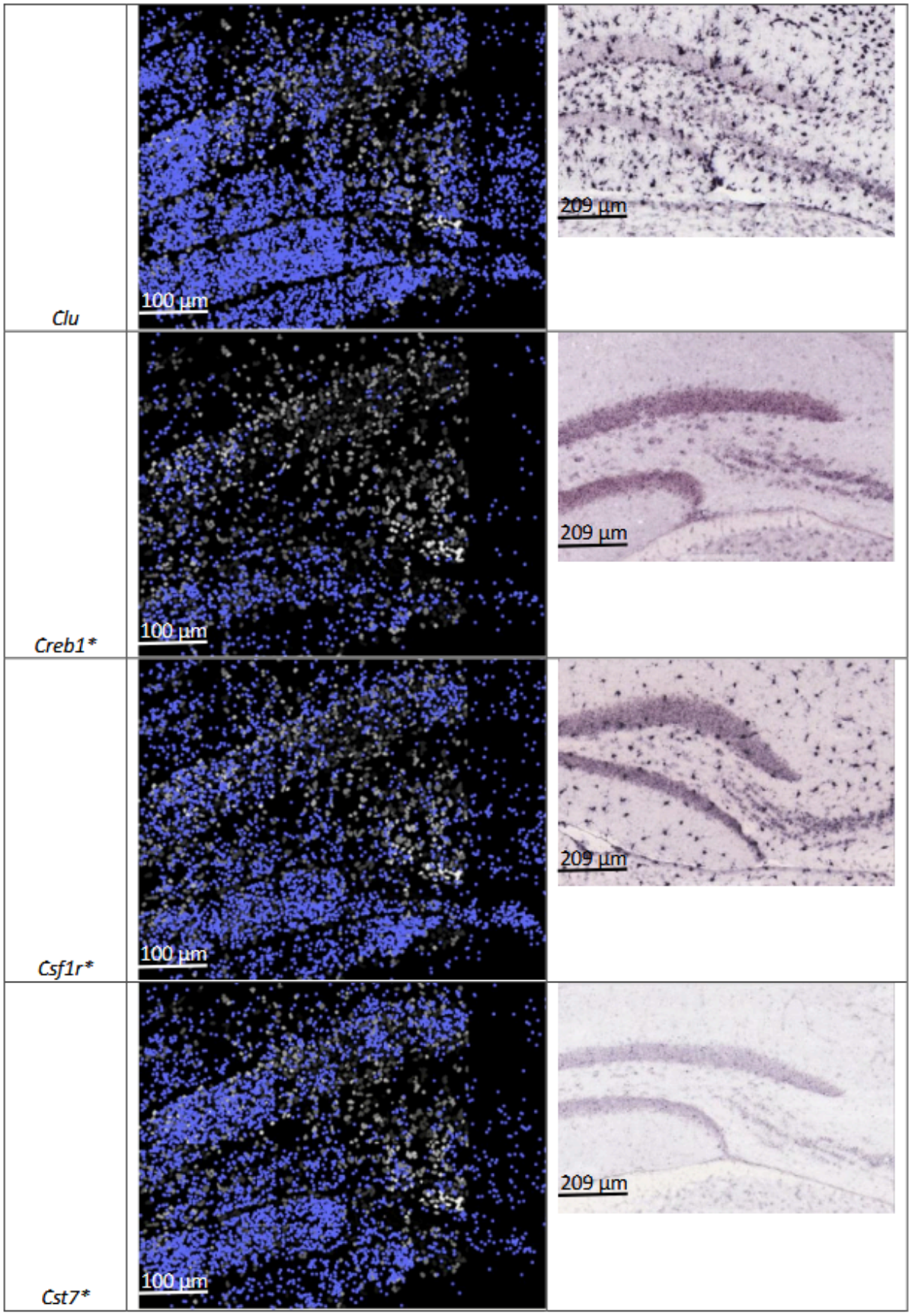
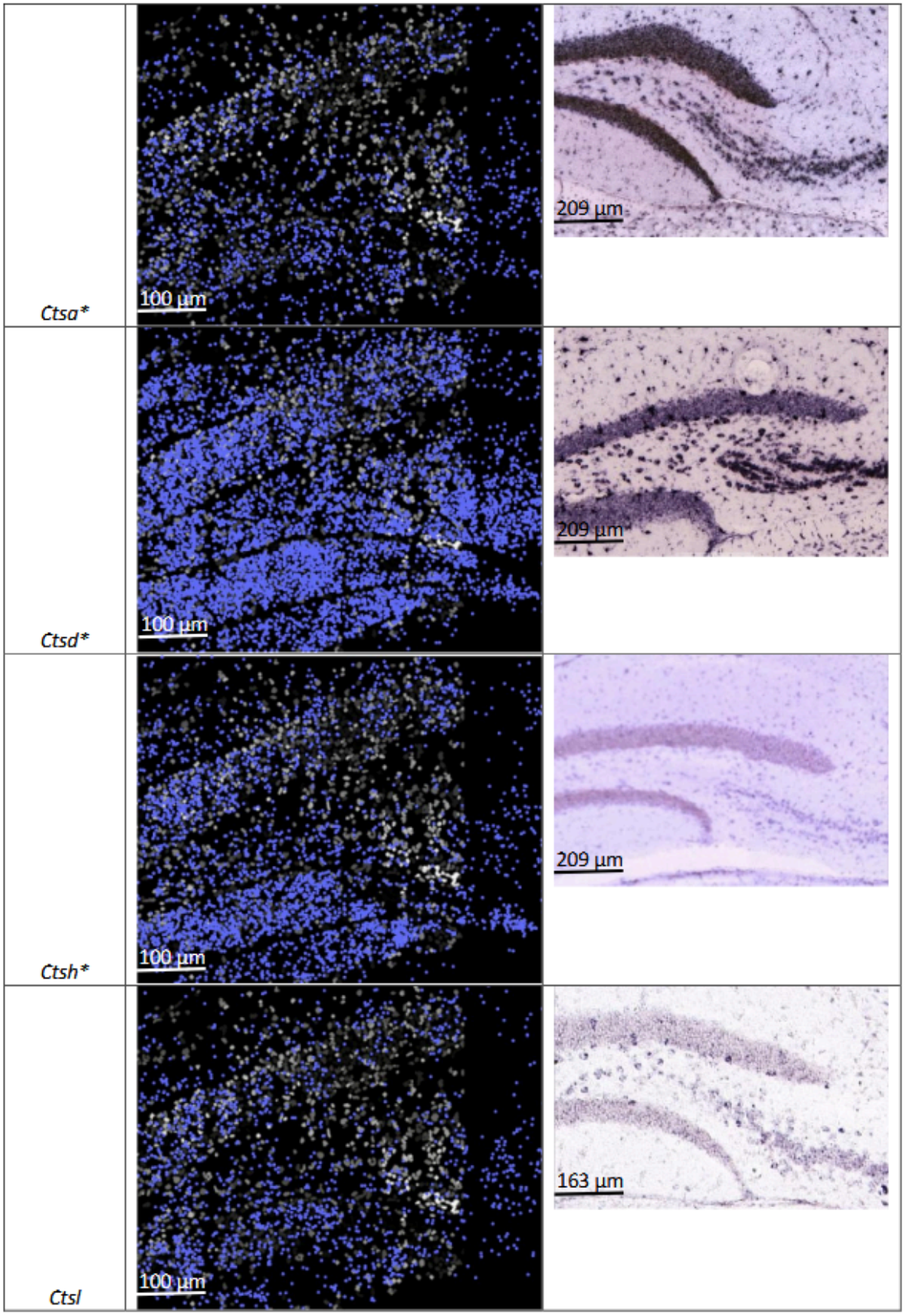
Gene	ExSeq	Allen Atlas
Actb		
*Apod**		
Apoe		
App		

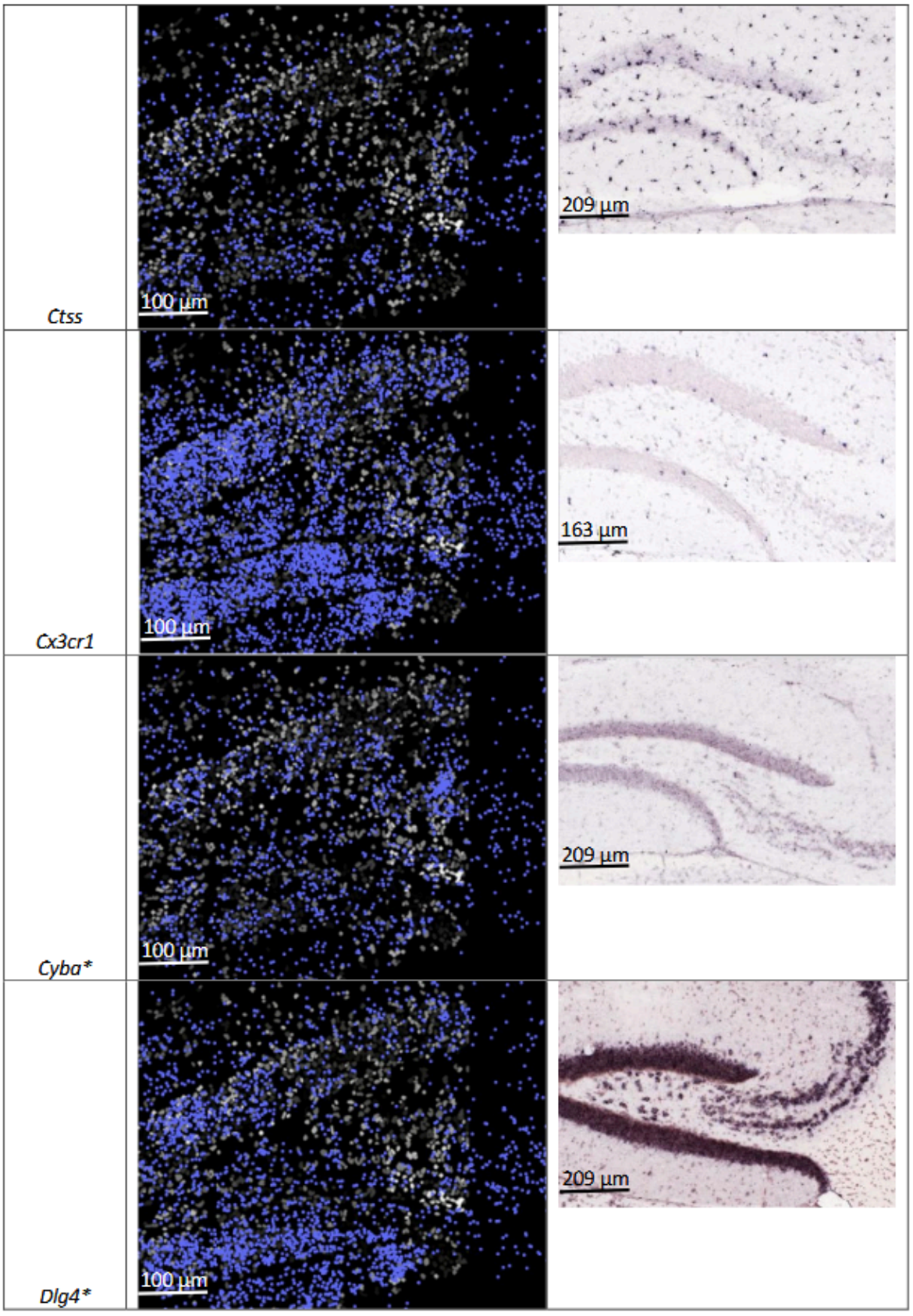

Aqp4
100 µm
163 µm
Baiap2l1*
100 µm
209 µm
Bsn
100 µm
209 µm
C1qa
100 µm
209 µm

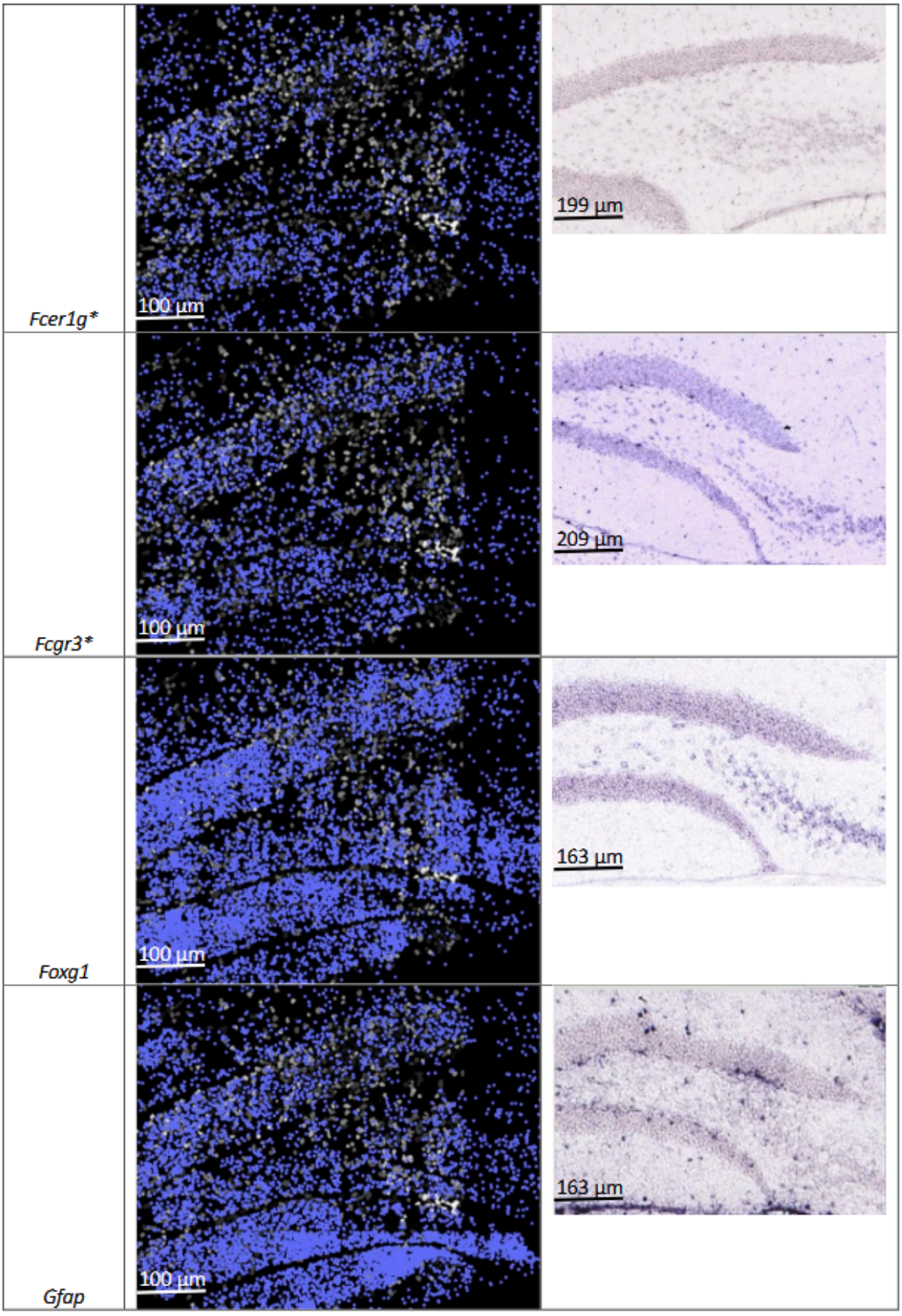

C1qc
100 µm
209 µm
C1ql2
100 µm
209 µm
C4a
100 µm
209 µm
Calb1
100 µm
209 µm

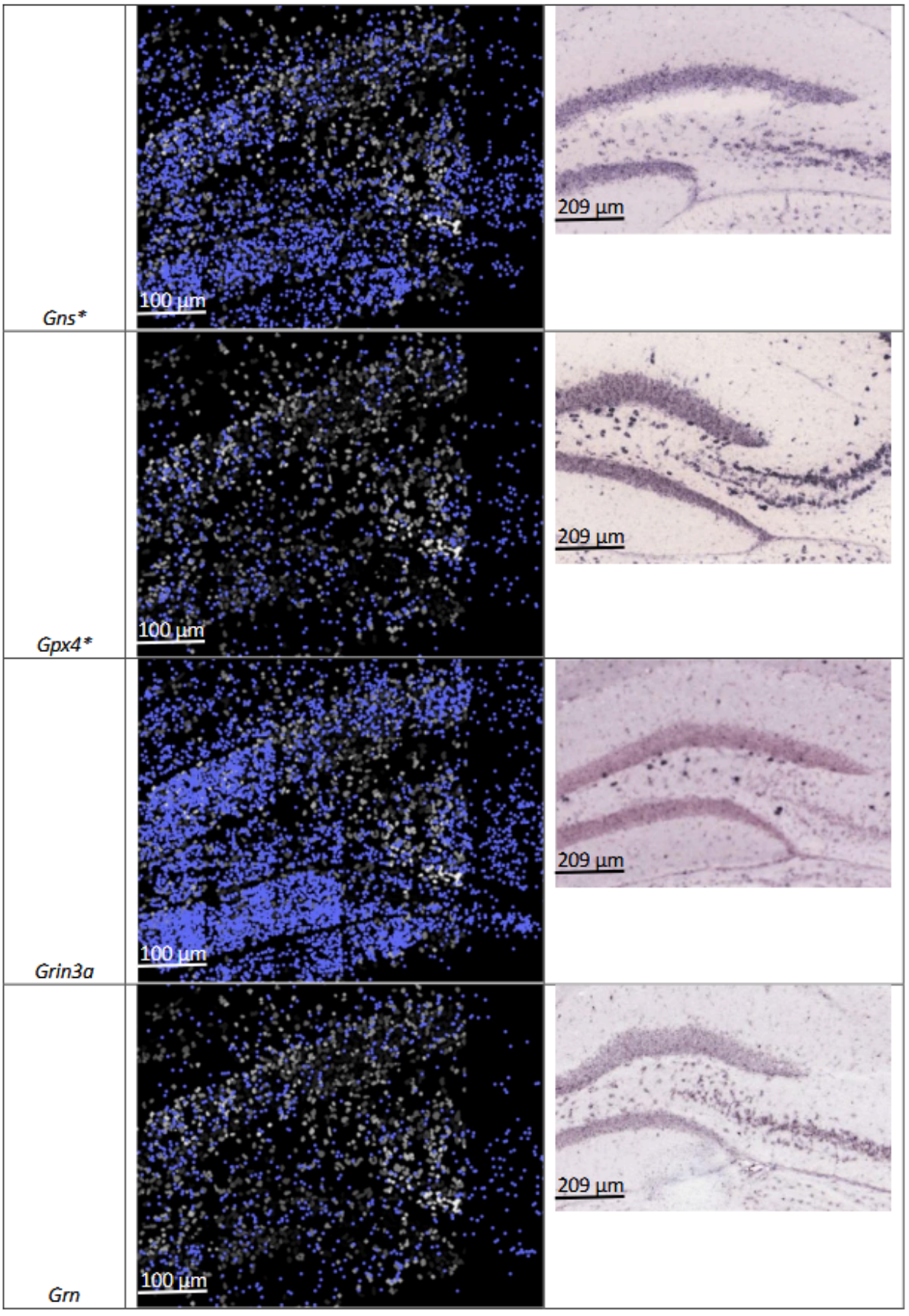

Casp8*
100 µm
209 µm
Ccrl2*
100 µm
209 µm
Cd68*
100 µm
209 µm
Cd9
100 µm
209 µm

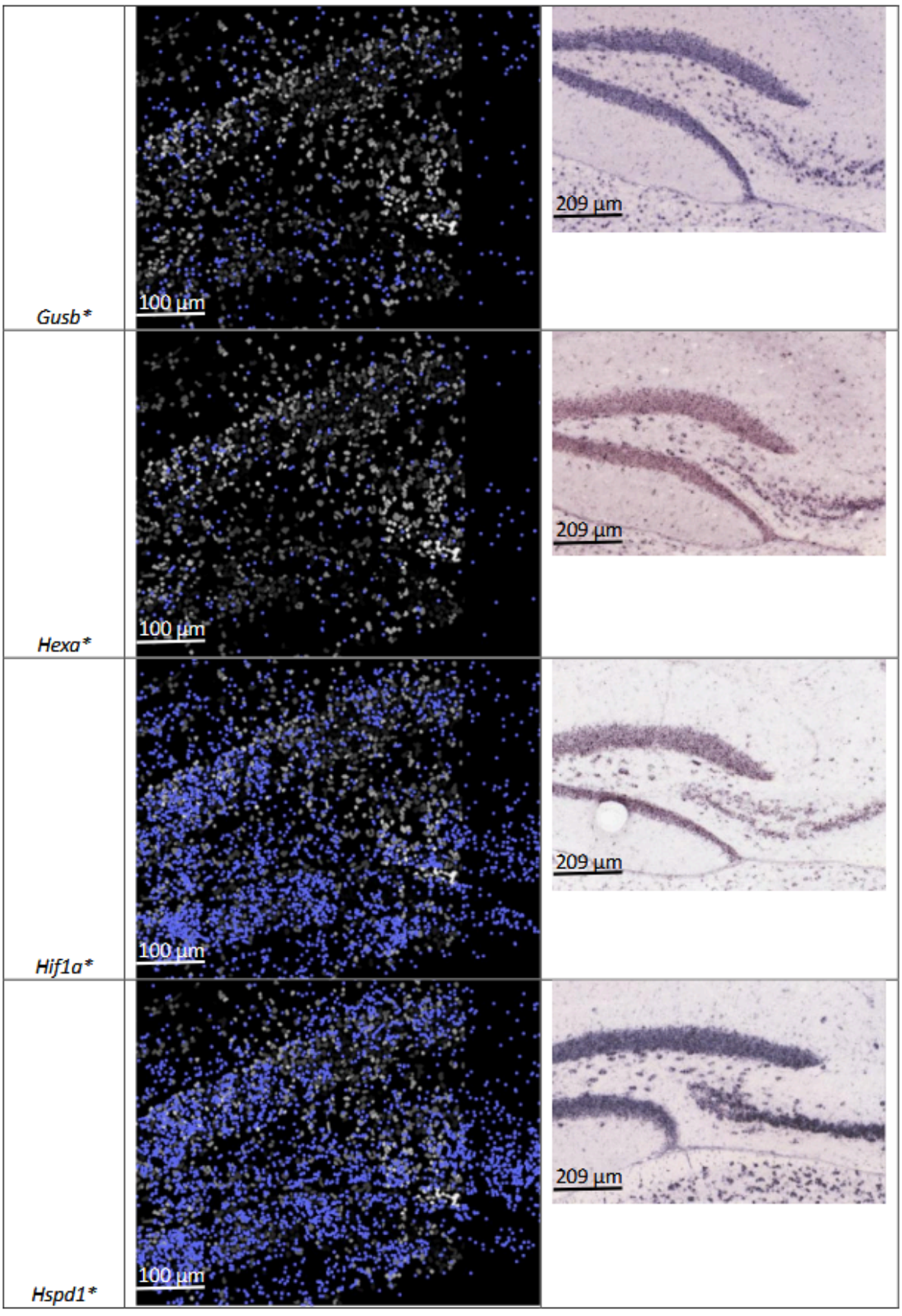

Clu
100 µm
209 µm
Creb1*
100 µm
209 µm
Csf1r*
100 µm
209 µm
Cst7*
100 µm
209 µm

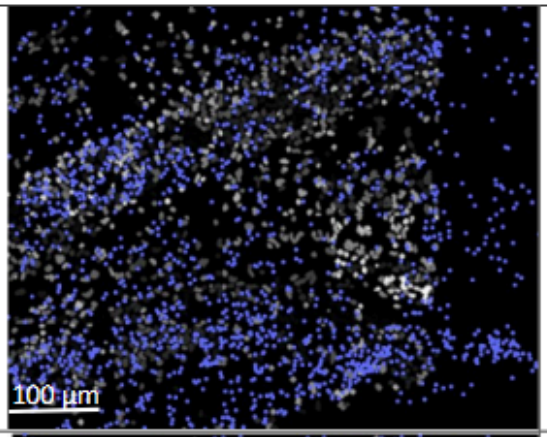
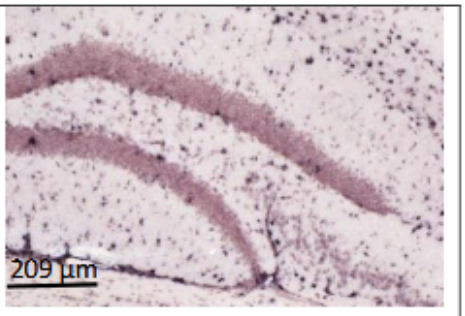
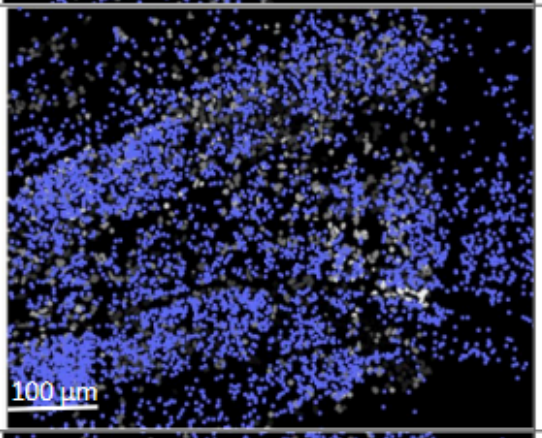
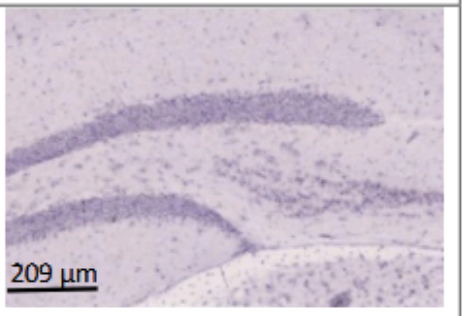
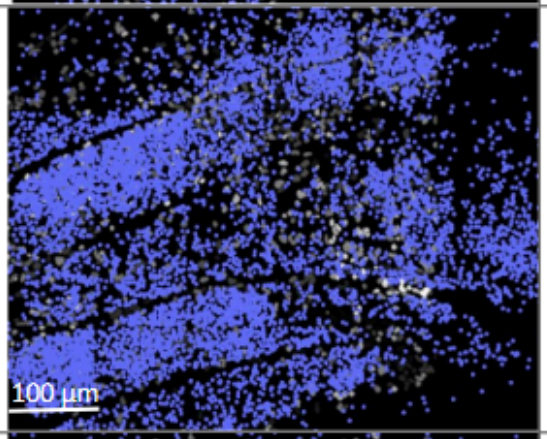
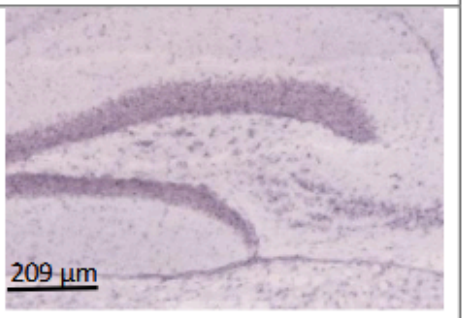
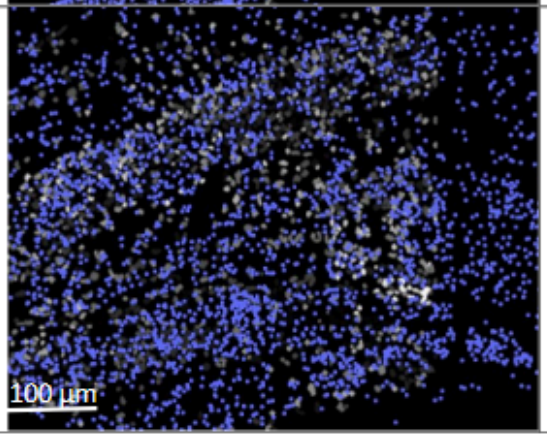
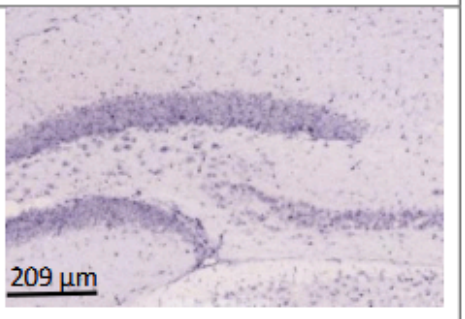

Ctsa*
100 µm
209 µm
Ctsd*
100 µm
209 µm
Ctsh*
100 µm
209 µm
Ctsl
100 µm
163 µm

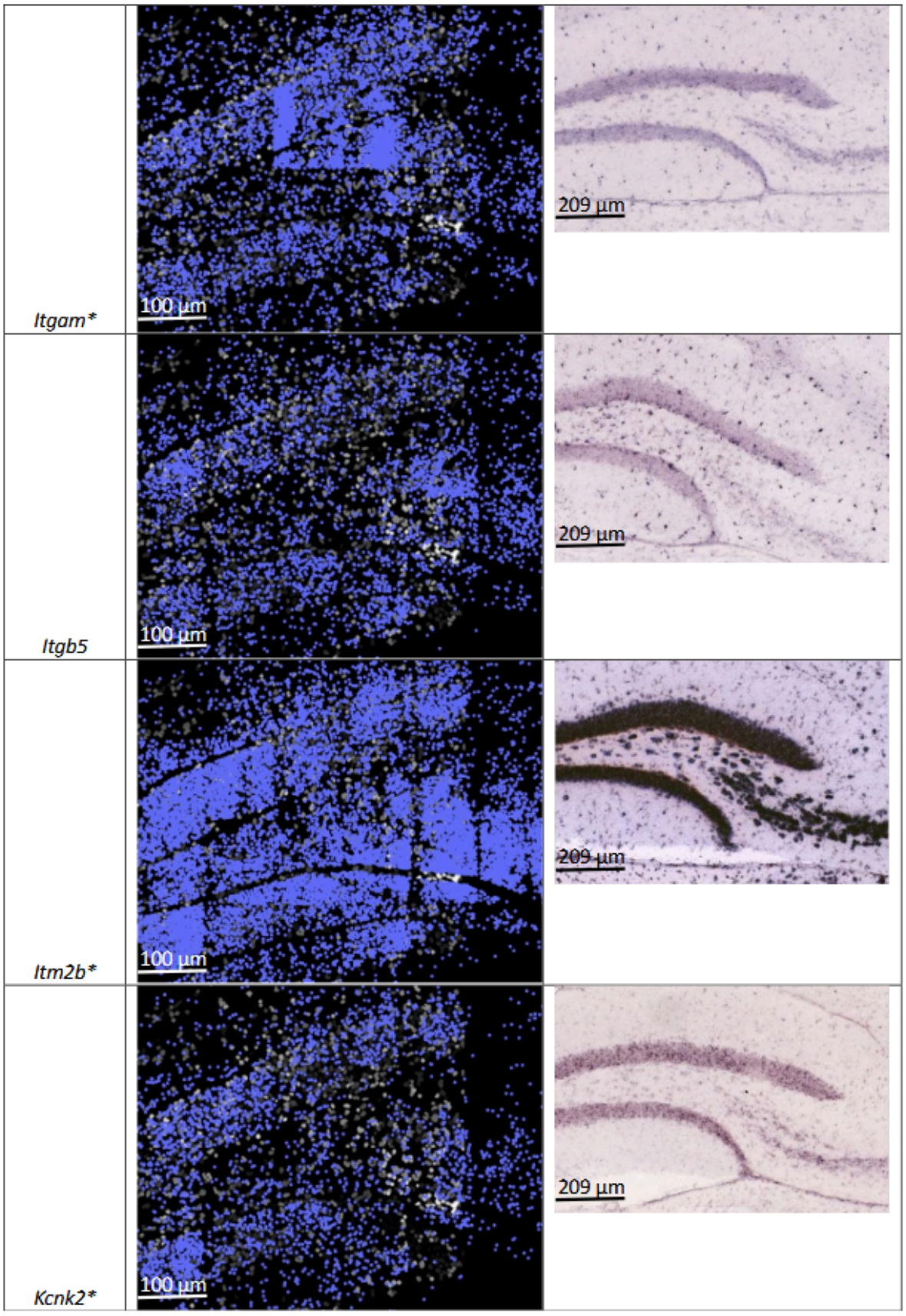

Ctss
100 µm
209 µm
Cx3cr1
100 µm
163 µm
Cyba*
100 µm
209 µm
Dlg4*
100 µm
209 µm

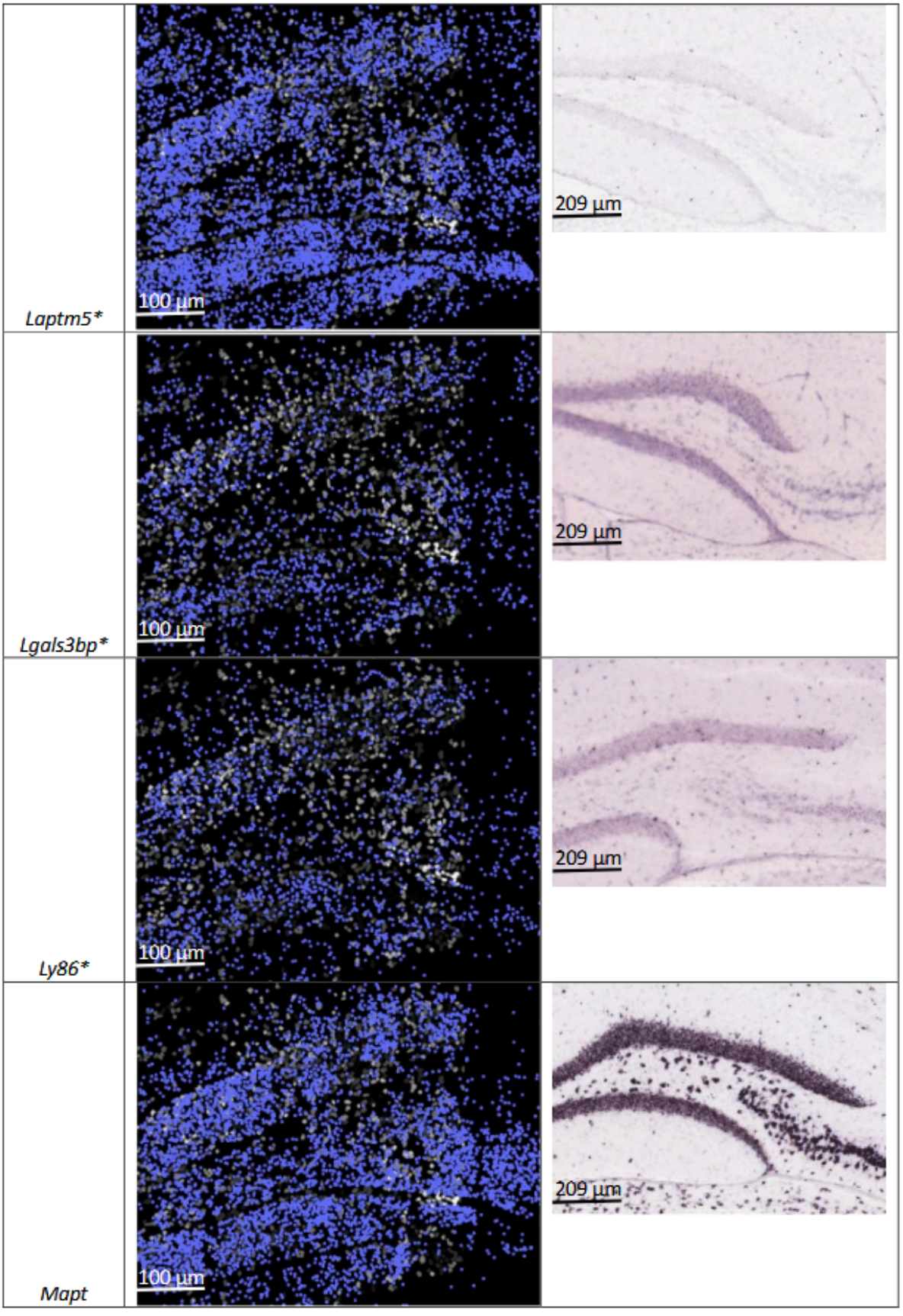

Fcer1g*
100 µm
199 µm
Fcgr3*
100 µm
209 µm
Foxg1
100 µm
163 µm
Gfap
100 µm
163 µm

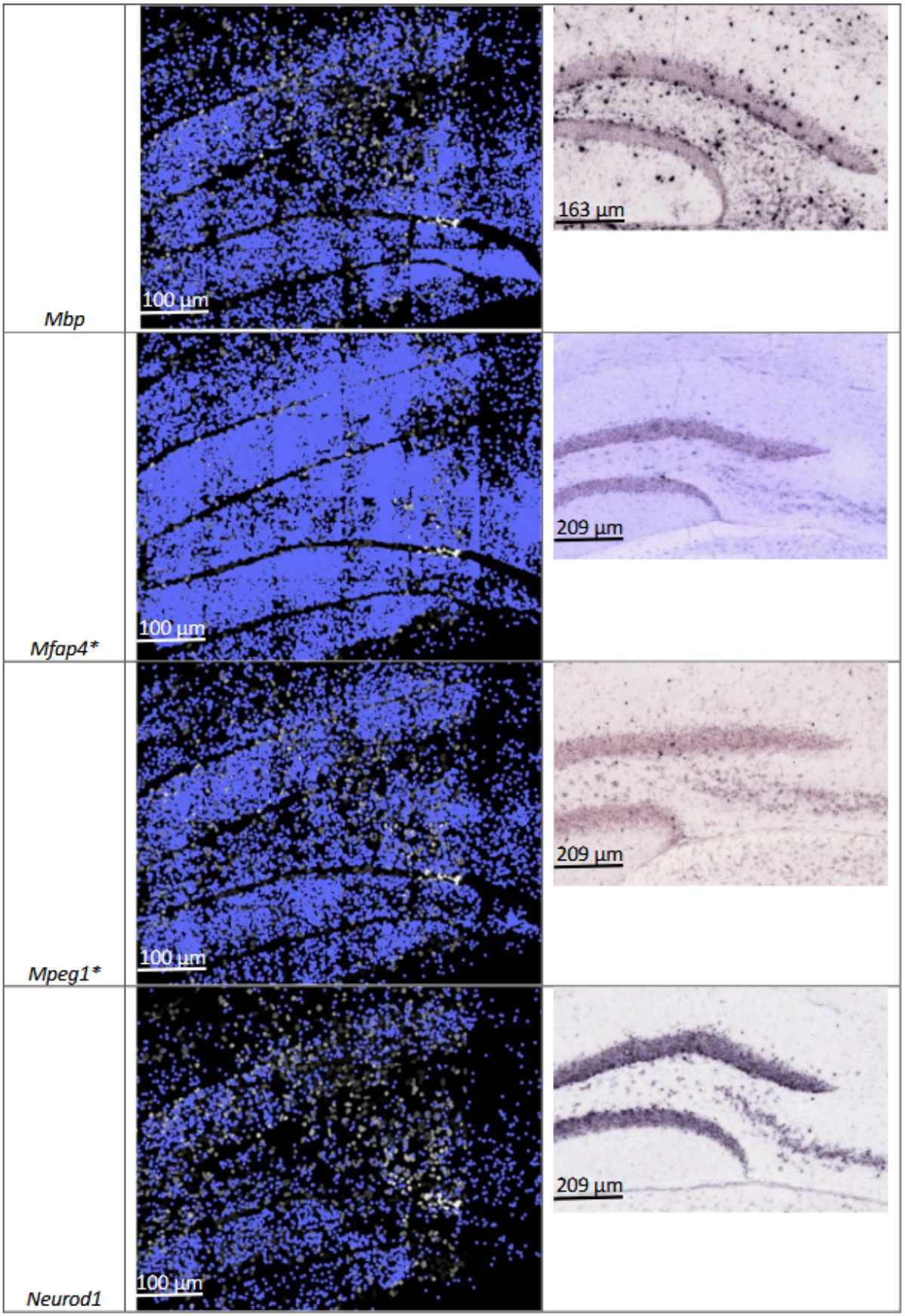

Gns*
100 µm
209 µm
Gpx4*
100 µm
209 µm
Grin3a
100 µm
209 µm
Grn
100 µm
209 µm

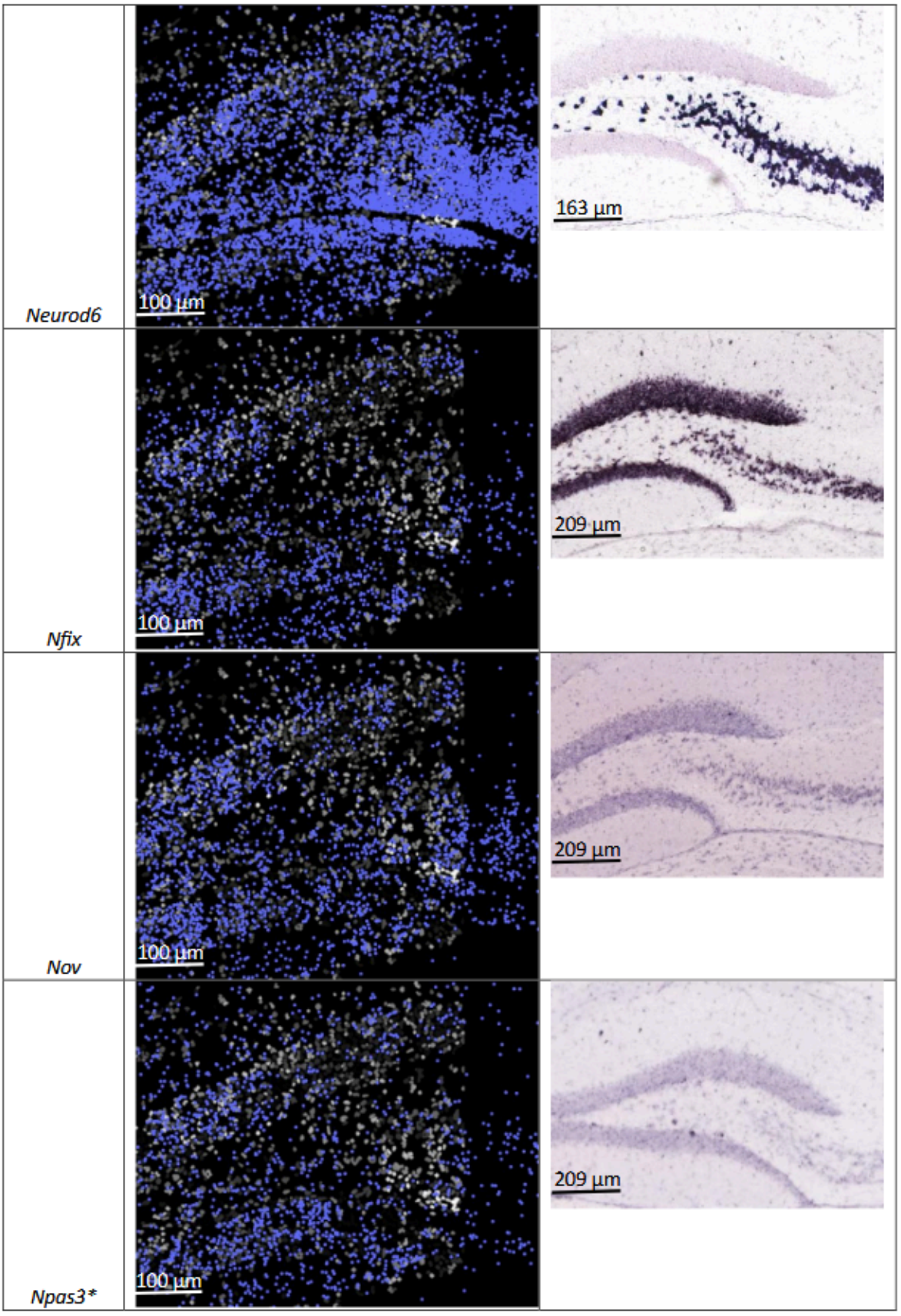

Gusb*
100 µm
209 µm
Hexa*
100 µm
209 µm
Hif1a*
100 µm
209 µm
Hspd1*
100 µm
209 µm

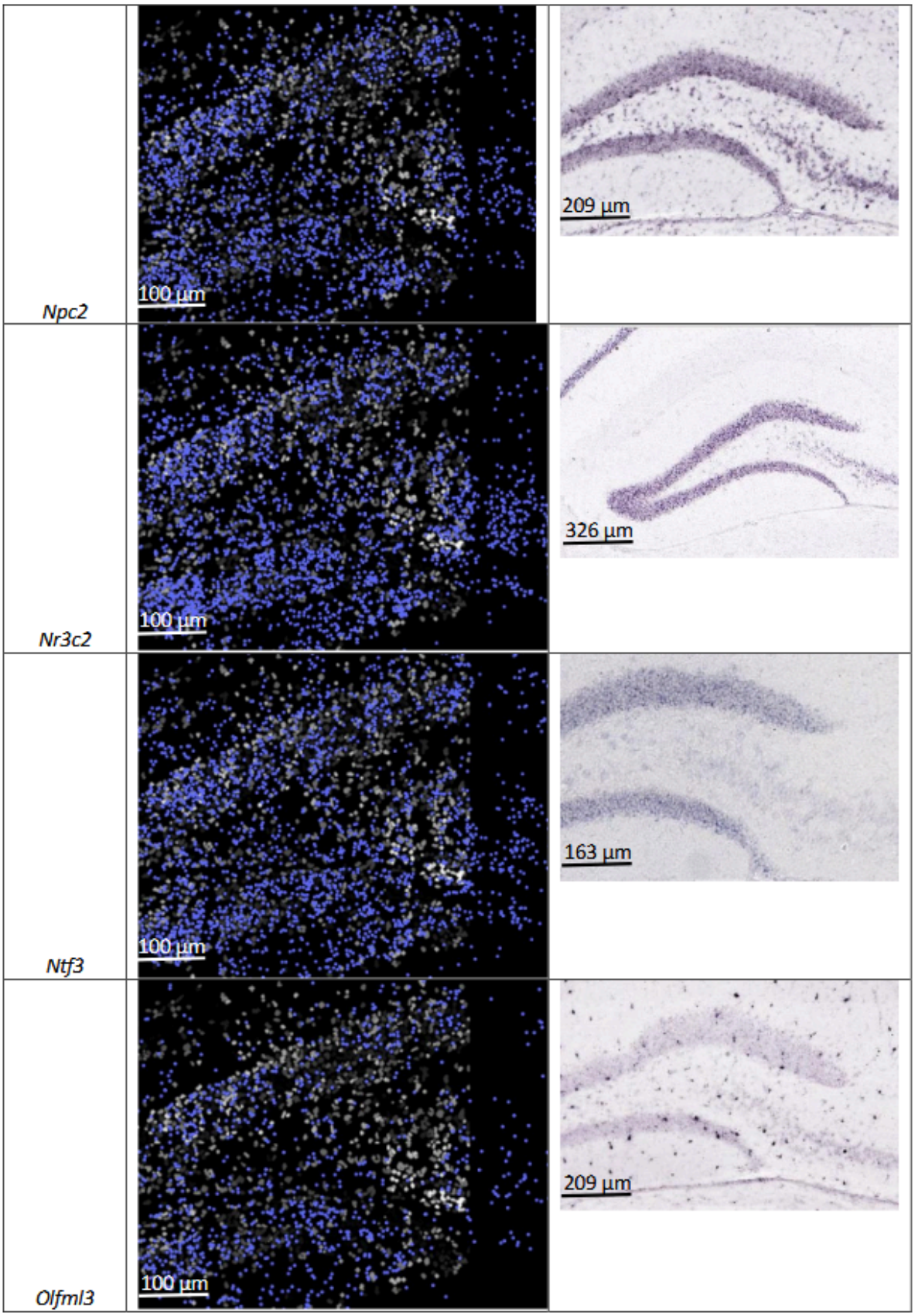
Idb3		
*inos**		
*Irg1**		
*Itga6**		

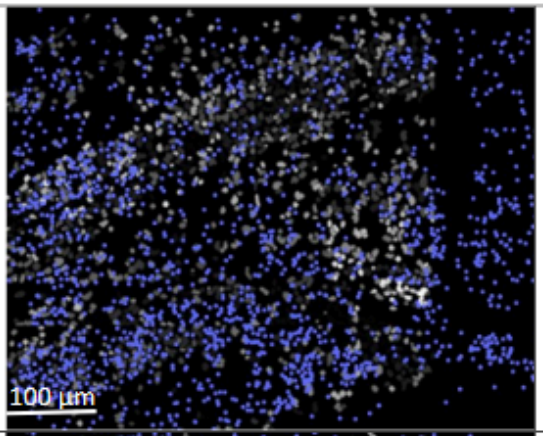
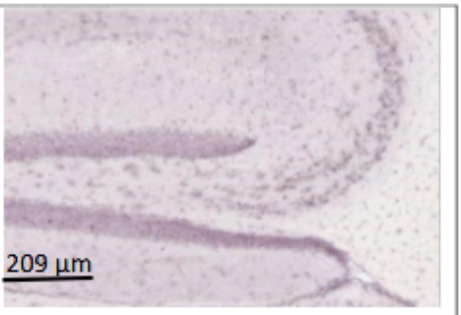
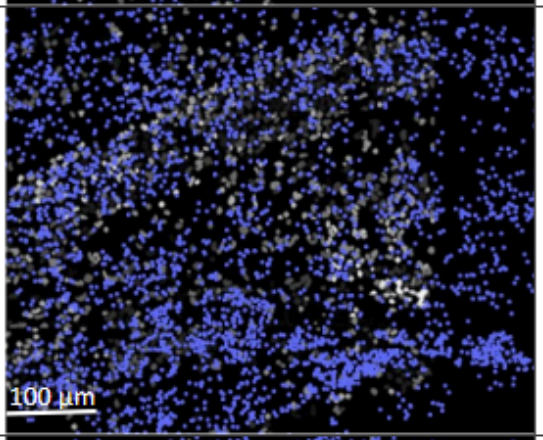
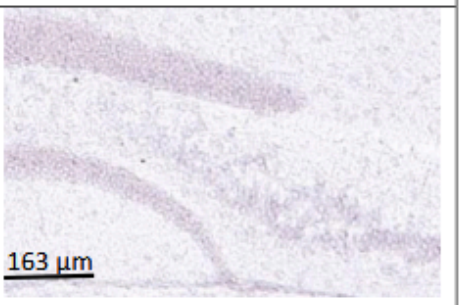
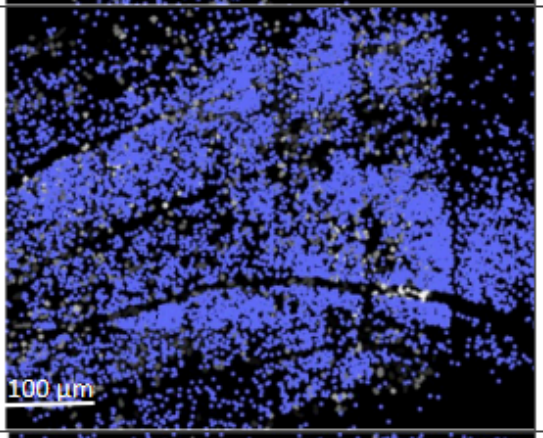
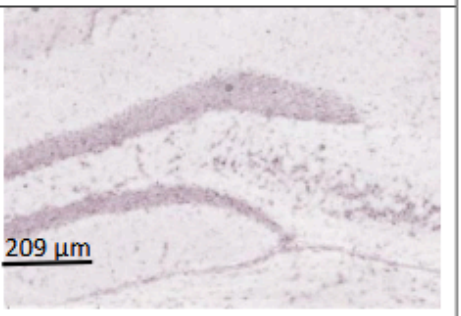
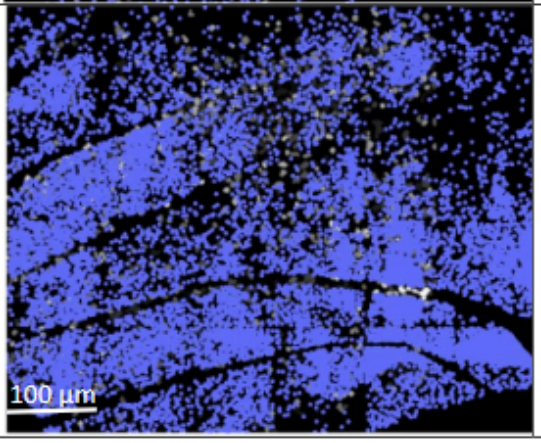
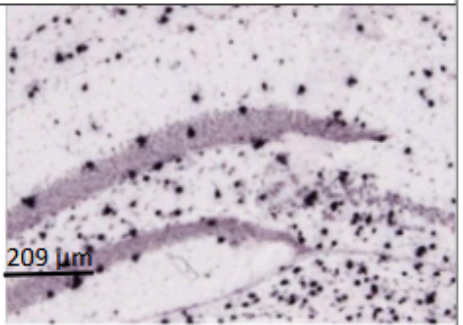

Itgam*
100 µm
209 µm
Itgb5
100 µm
209 µm
Itm2b*
100 µm
209 µm
Kcnk2*
100 µm
209 µm

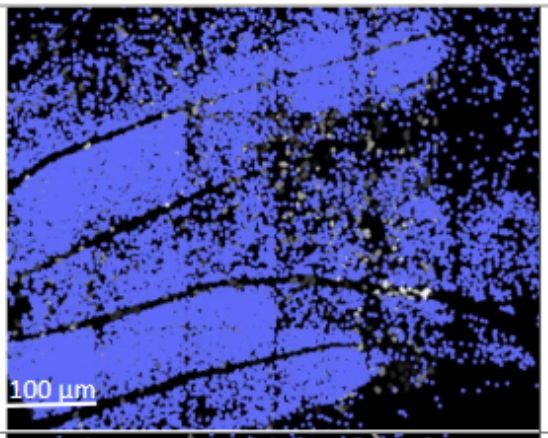
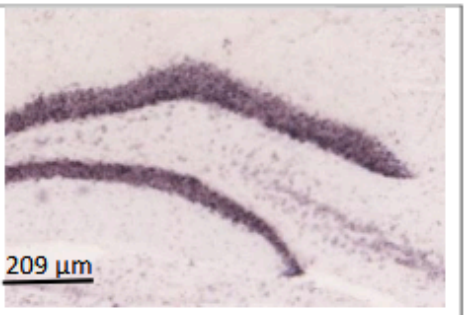
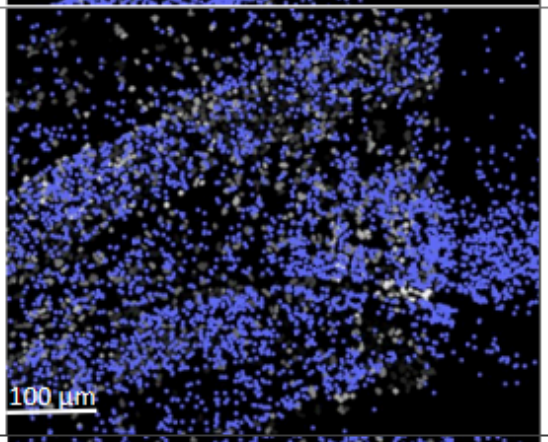
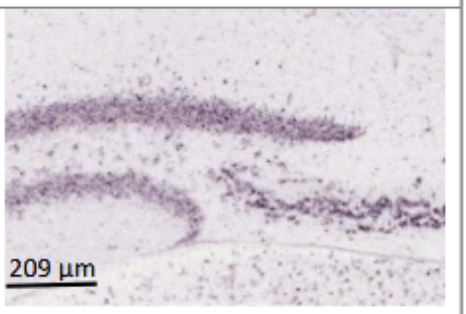
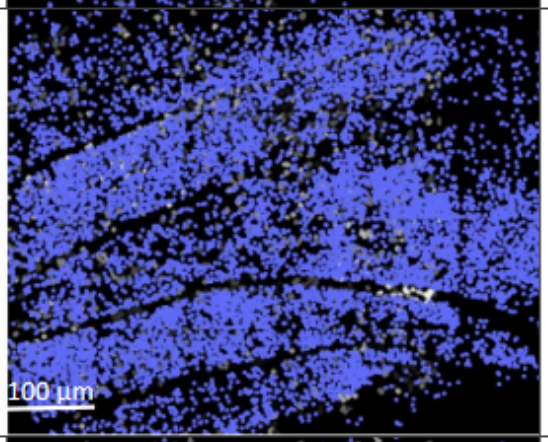
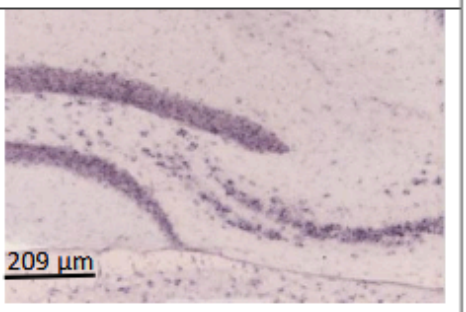
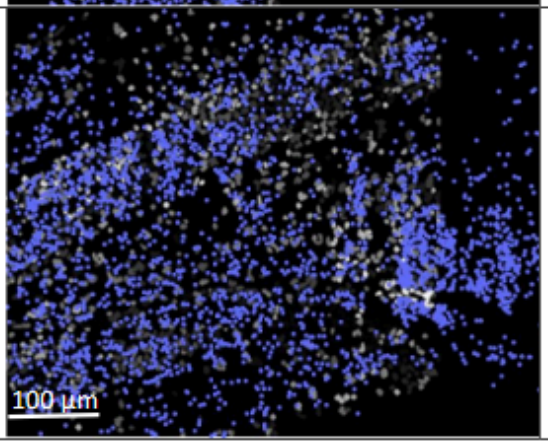
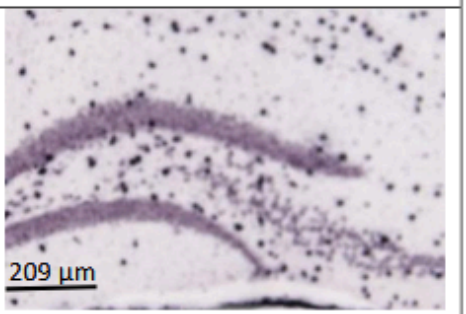

Laptm5*
100 µm
209 µm
Lgals3bp*
100 µm
209 µm
Ly86*
100 µm
209 µm
Mapt
100 µm
209 µm

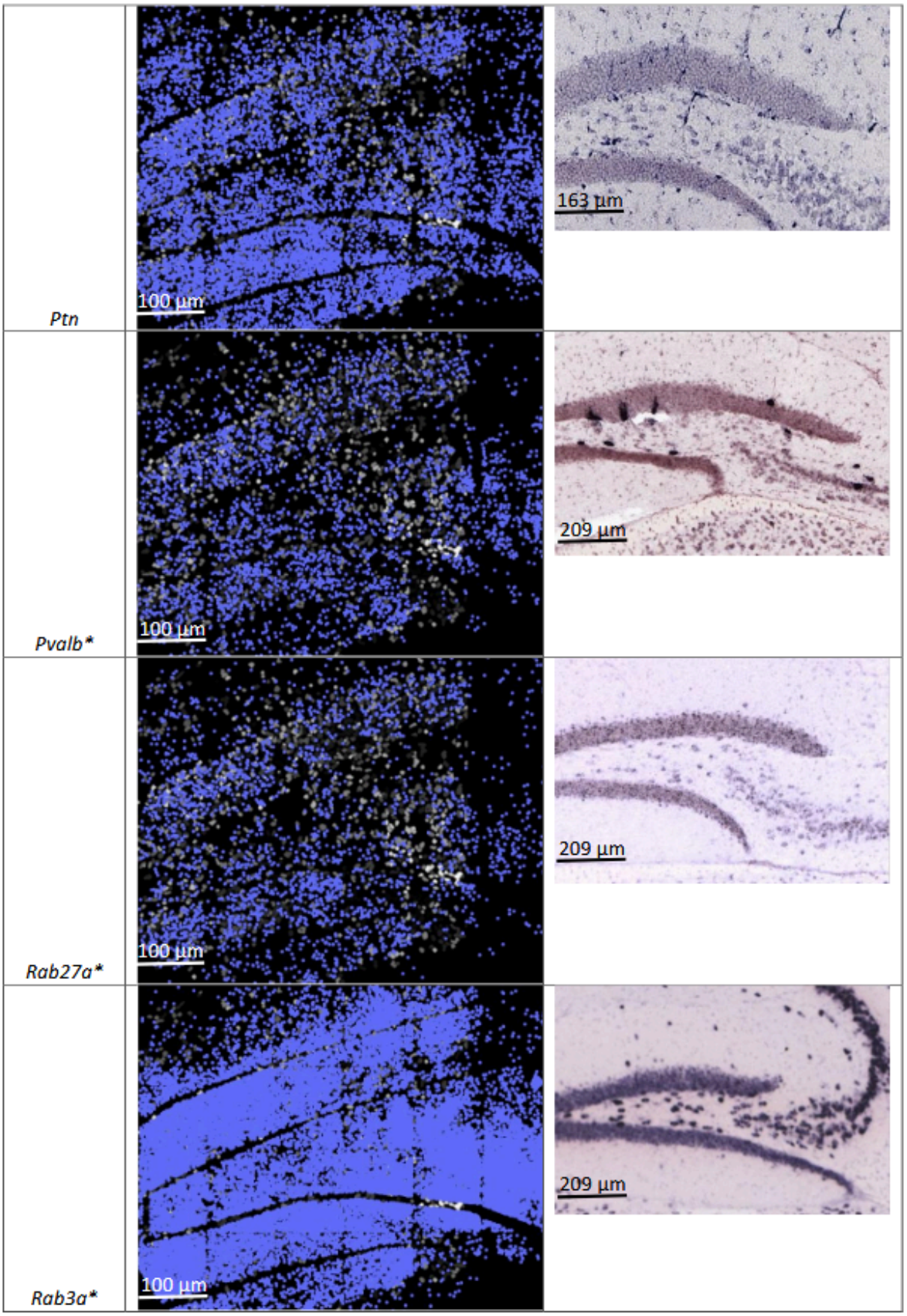

Mbp
100 µm
163 µm
Mfap4*
100 µm
209 µm
Mpeg1*
100 µm
209 µm
Neurod1
100 µm
209 µm

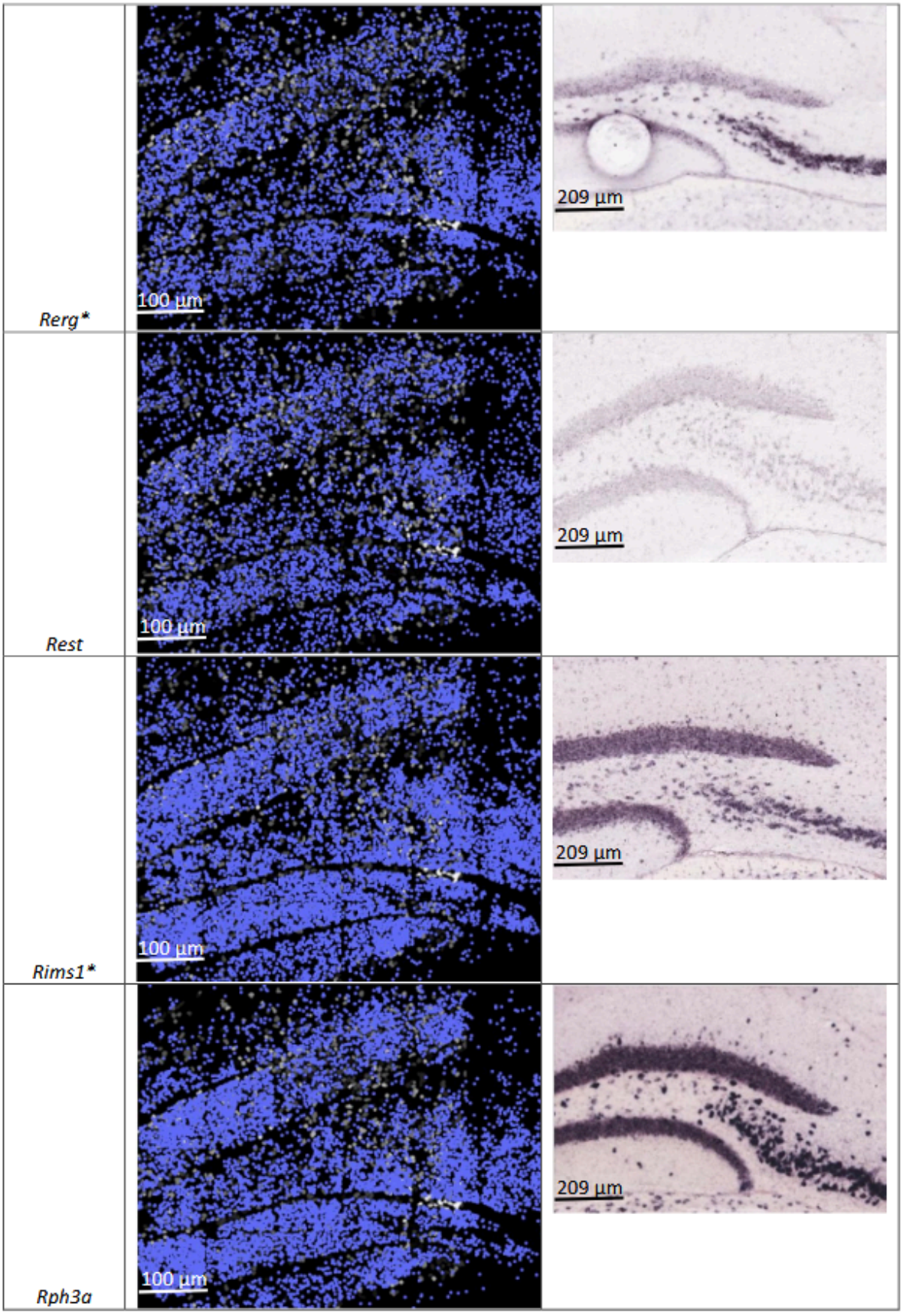

Neurod6
100 µm
163 µm
Nfix
100 µm
209 µm
Nov
100 µm
209 µm
Npas3*
100 µm
209 µm

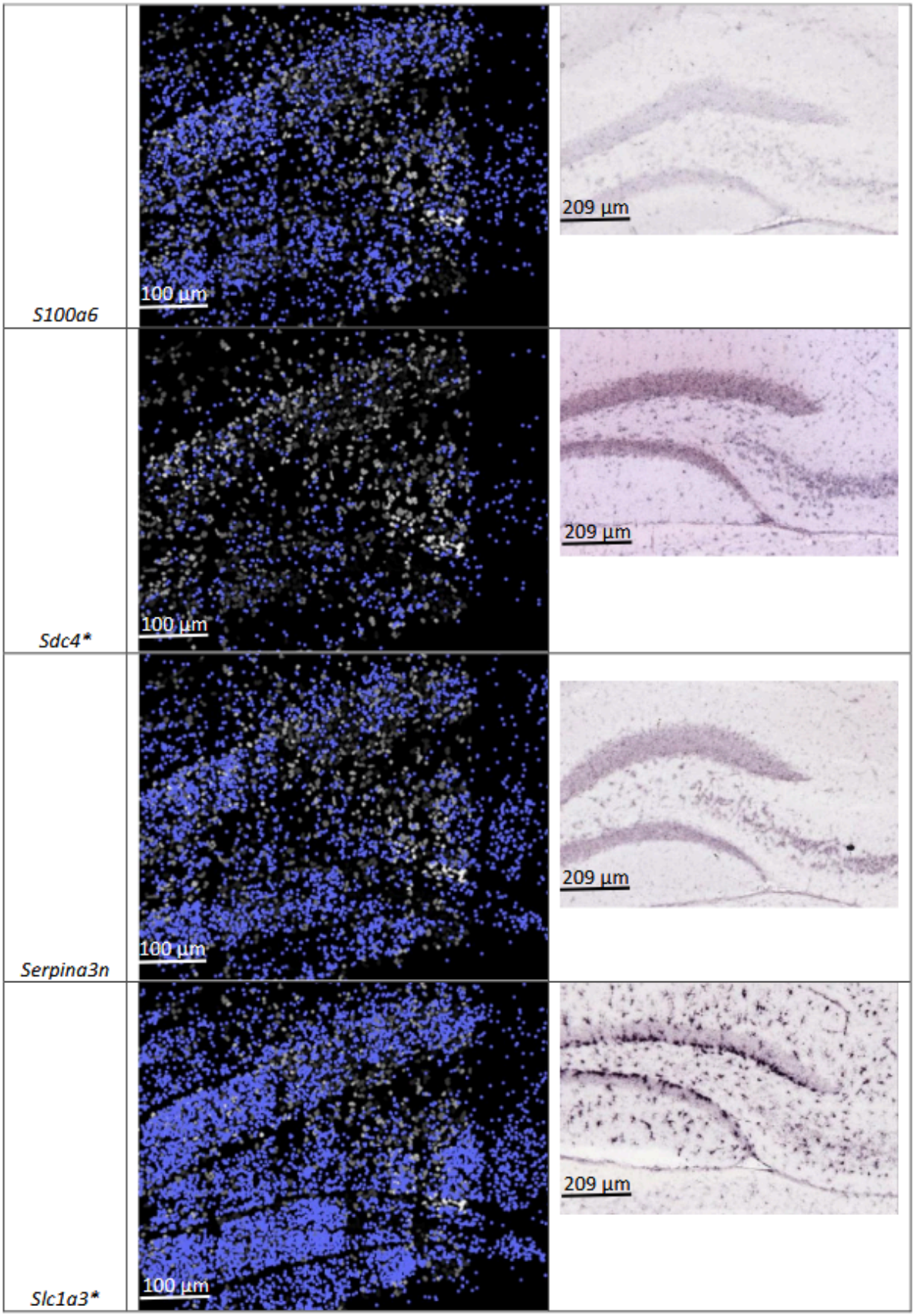

Npc2
100 µm
209 µm
Nr3c2
100 µm
326 µm
Ntf3
100 µm
163 µm
Olfml3
100 µm
209 µm

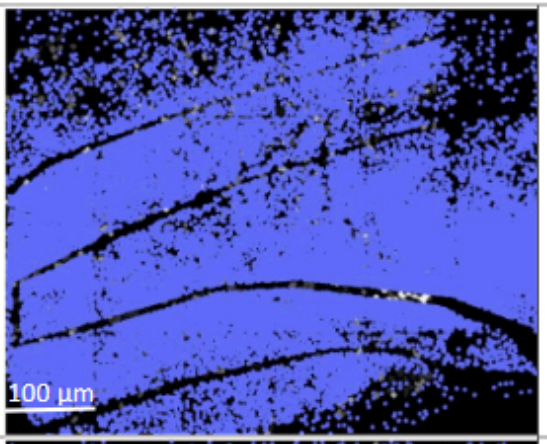
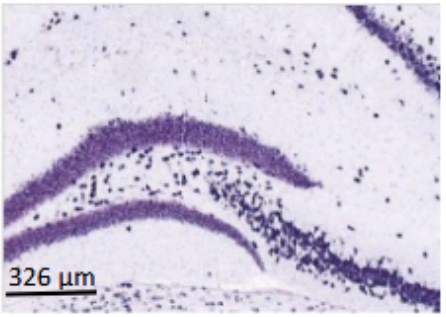
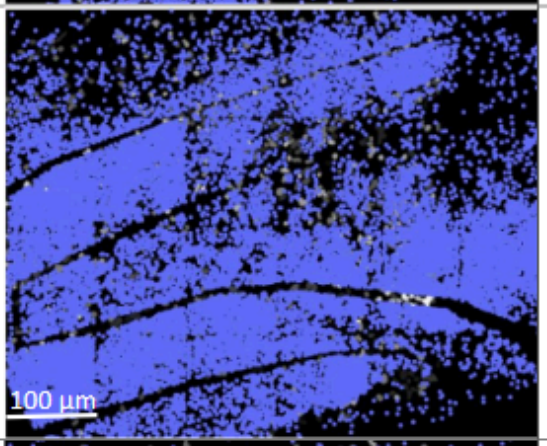
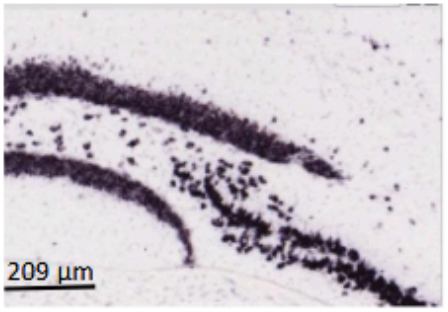
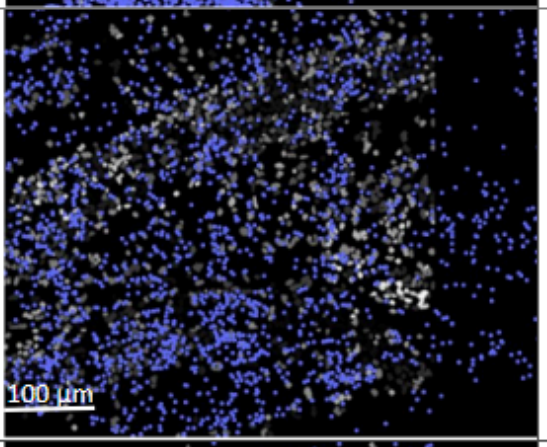
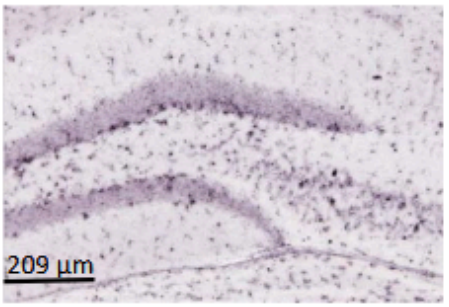
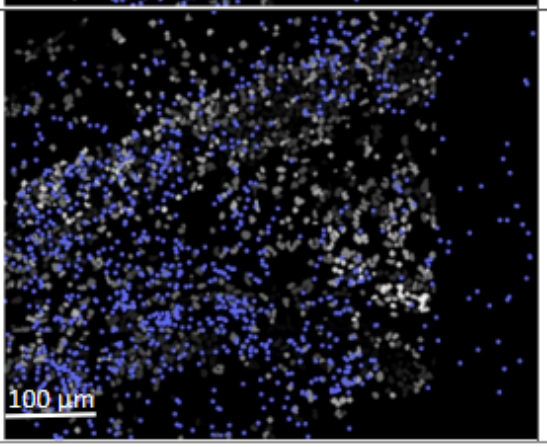
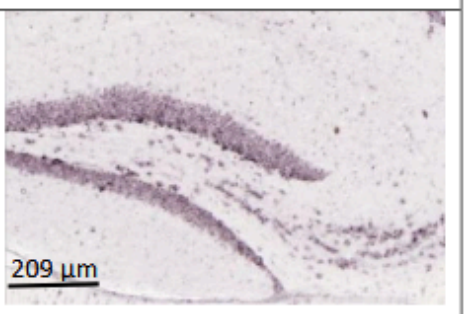
*P2ry1**		
Pax6		
Plek		
Plp1		

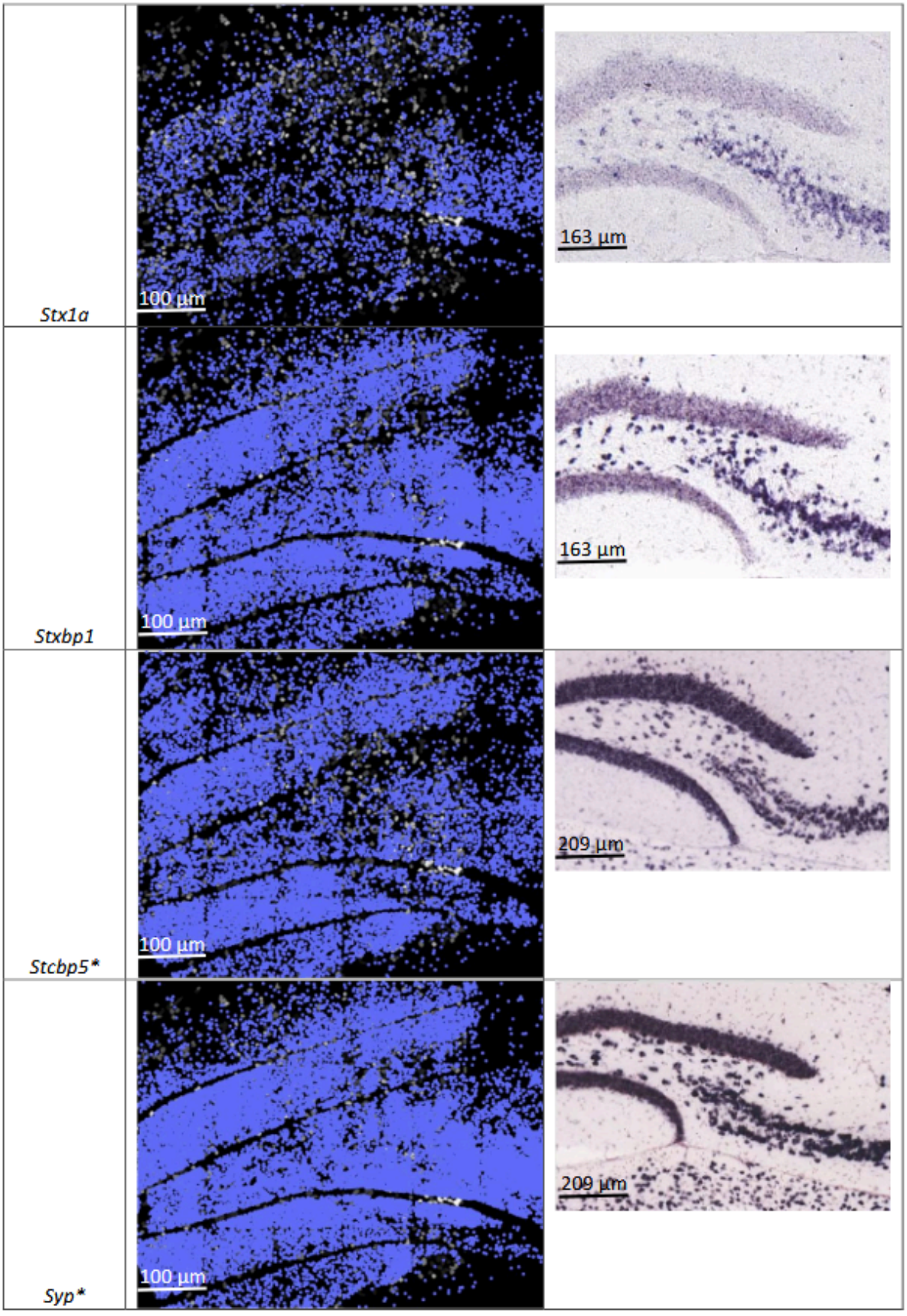
Prox1		
*Psen1**		
*Psen2**		
Ptgds		

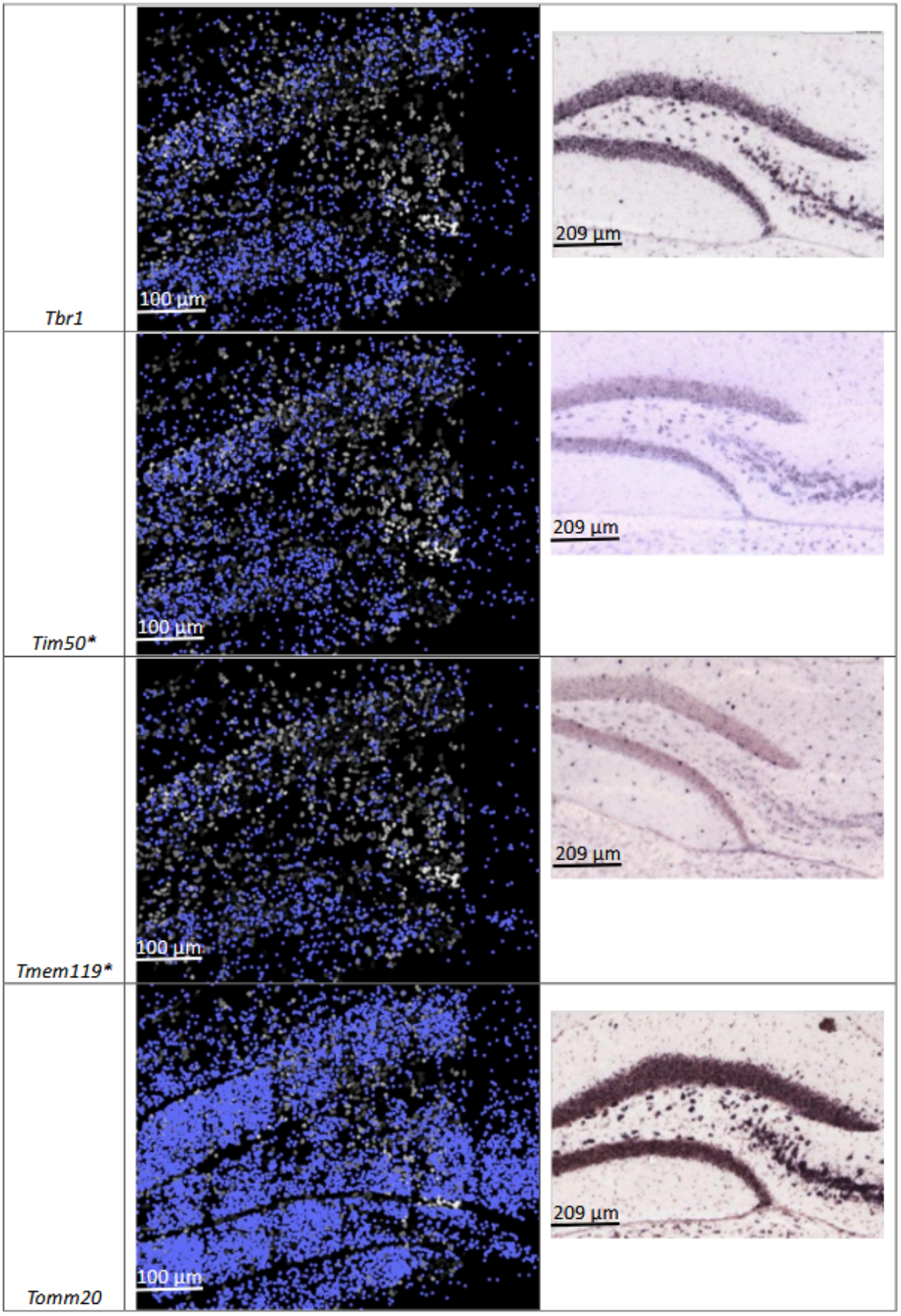

Ptn
100 µm
163 µm
Pvalb*
100 µm
209 µm
Rab27a*
100 µm
209 µm
Rab3a*
100 µm
209 µm

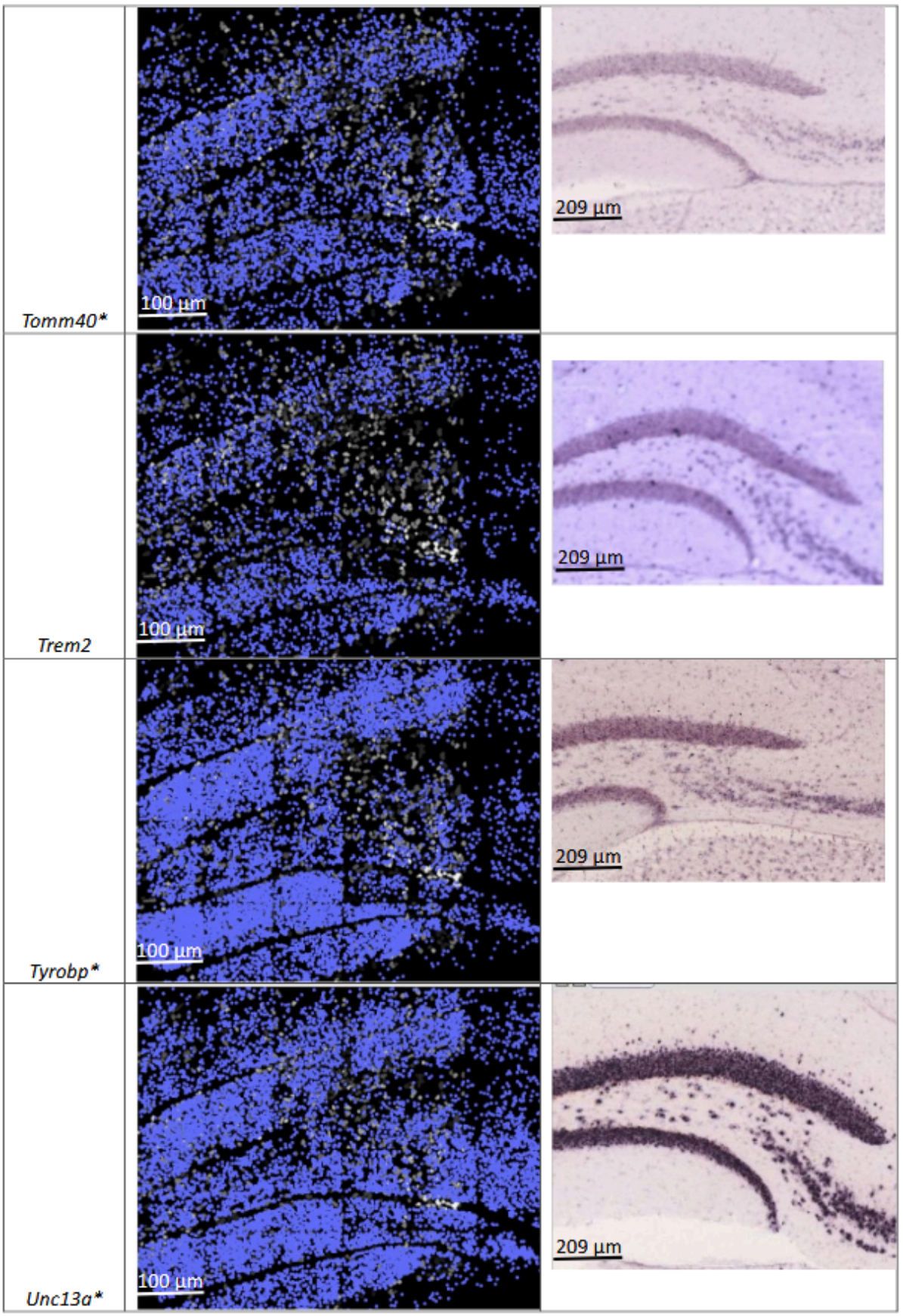

Rerg*
100 µm
209 µm
Rest
100 µm
209 µm
Rims1*
100 µm
209 µm
Rph3a
100 µm
209 µm


S100a6
100 µm
209 µm
Sdc4*
100 µm
209 µm
Serpina3n
100 µm
209 µm
Slc1a3*
100 µm
209 µm

Snap25		
Snca		
Sox2		
*Sox4**		


100 µm
163 µm
Stx1a
100 µm
163 µm
Stxbp1
100 µm
209 µm
Stcbp5*
100 µm
209 µm
Syp*


Tbr1
100 µm
209 µm
Tim50*
100 µm
209 µm
Tmem119*
100 µm
209 µm
Tomm20
100 µm
209 µm


Tomm40*
100 µm
209 µm
Trem2
100 µm
209 µm
Tyrobp*
100 µm
209 µm
Unc13a*
100 µm
209 µm

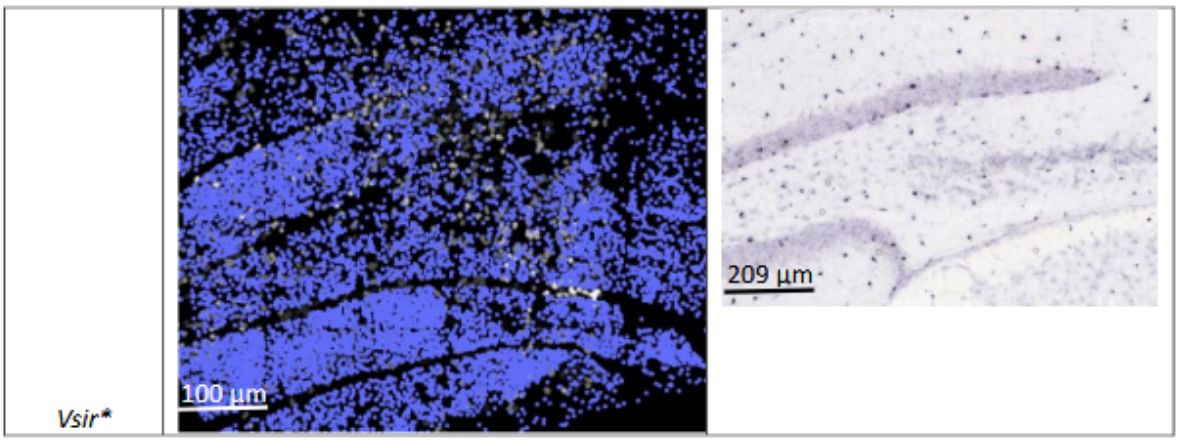


Fig. S2. Comparison of ExSeq data with the Allen Mouse Brain Atlas. Spatial expression patterns of individual genes in the WT sample WT_2 were qualitatively compared to in situ hybridization data from the Allen Mouse Brain Atlas. ExSeq plots show DAPI-stained nuclei (gray) and gene locations (blue). In the Allen Brain Atlas purple indicates high gene expression in FISH experiments. Genes marked with an asterisk (*) were compared to sagittal sections when coronal sections were unavailable in the Atlas; in these cases, the Atlas images were horizontally flipped to approximate anatomical orientation. Overall, the observed spatial distributions showed good agreement with Allen Atlas reference patterns, supporting the validity of the ExSeq data.